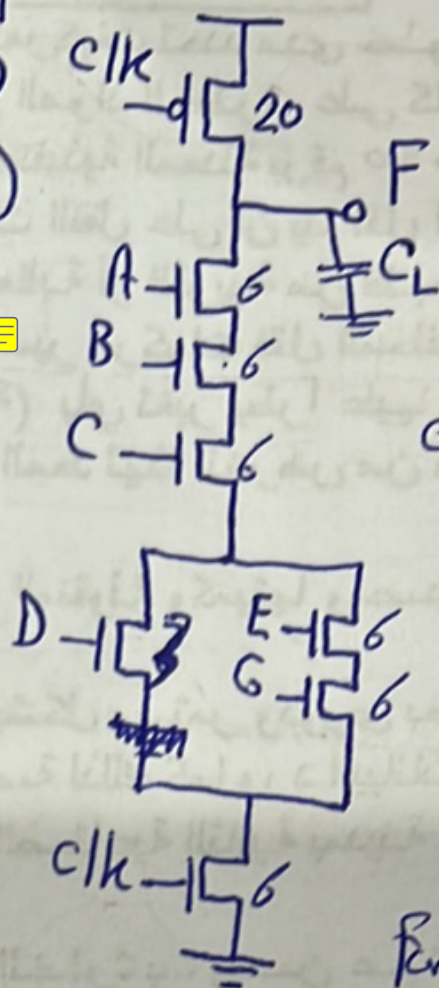
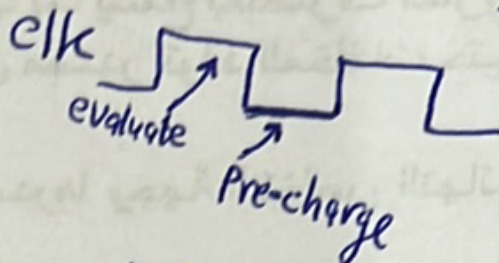




c) $Q_L = 26 C_0$

$$d) \mathcal{S} = \frac{T_h}{T_h + T_c}$$



$$T_L \geq 0.69 \frac{R_o}{f_o} \cdot C_L = 0.69 \frac{R_o}{f_o} 260$$

$$T_h \geq 0.69 R_o C_L = 0.69 R_o 26 C_o$$

for max frequency :

use $T_e = 0.69 \frac{R_o}{10} 26 C_o$

$$T_h = 0.69 R_{0.26} C_o$$

$$\therefore S = 90.91\%$$

$$e) f_{\max} = \frac{1}{T_{\text{clk}, \min}} = \frac{1}{T_{\text{L}, \min} + T_{\text{H}, \min}}$$

$$\epsilon_0 f_{\max} = \frac{1}{0.69 (1.1 R_0) (26 c_0)}$$

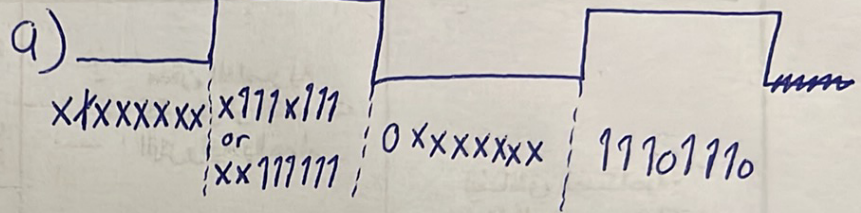
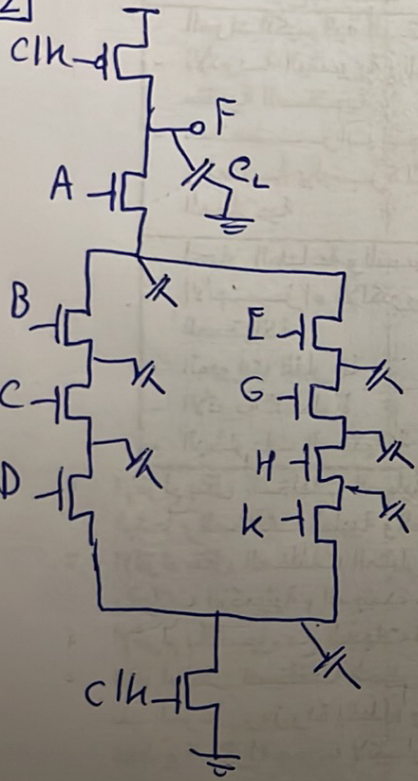
$$f) I_{\text{leak}} = C_L \frac{\Delta V}{\Delta t} \quad \therefore \Delta t_{\text{max}} = \frac{C_{\text{out}} \Delta V_{\text{out,max}}}{I_{\text{leak}}} = \frac{100 \text{ pF} \times 0.1 \times 3}{1 \text{ nA}} = 30 \text{ ms}$$

∴ $T_{h, \max} = 30 \text{ ms}$, assume $T_L = 0.69 \frac{R_o}{10} \cancel{\text{MVA}} C_L = 0.69 \frac{R_o}{10} (100 \text{ pF}) = 0.69 \times \frac{1}{10} \times 100 \times 10^{-12} R_o$

$$f_{\min} = \frac{1}{30 \text{ ms} + 0.69 \times \frac{1}{10} \times 100 \times 10^{-12} \text{ } R_o}$$

~~$\frac{S}{P} \times \frac{R}{P} = \frac{R}{P}$~~

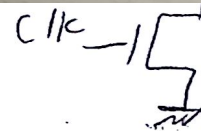
2



b) $C_i < \frac{1}{10} C_L$

$$V_{out} = \frac{C_1 V_1 + C_2 V_2 + \dots}{C_1 + C_2 + \dots} = \frac{C_L V_{DD}}{C_L + 6 C_i} = \frac{C_L V_{DD}}{C_L + 6 \times \frac{1}{10} C_L}$$

$V_{out, worst} = 0.8 V_{DD}$

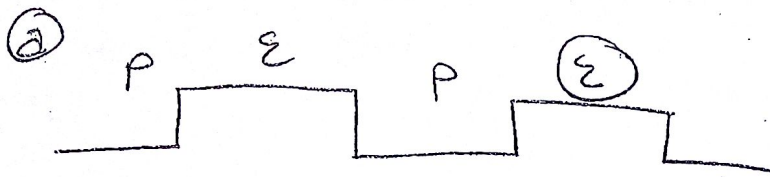


~~$V_{out} = \frac{C_L V_{DD}}{C_L + 6 C_i} = \frac{C_L V_{DD}}{C_L + 6 \times \frac{1}{10} C_L} = 0.8 V_{DD}$~~

~~$V_{out} = \frac{C_L V_{DD}}{C_L + 6 C_i} = \frac{C_L V_{DD}}{C_L + 6 \times \frac{1}{10} C_L} = 0.8 V_{DD}$~~

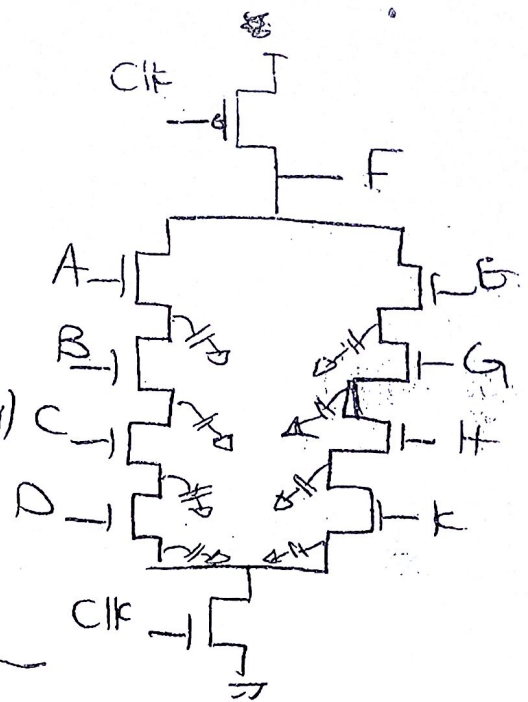
3) $F = ABCD + EGHK$

$$C_X = \frac{5}{100} C_L$$



1st P of second evaluate (no charge sharing)

A	B	C	D	E	G	H	K
0	X	X	X	0	X	X	X



b)

$$C_L U_{DD} = (C_L + 6C_X) U_X$$

$$V_X = 0.76 U_{DD}$$

c)



second precharge (no charge sharing)

A	B	C	D	E	G	H	K
1	1	1	X	1	1	1	X

d)



first precharge (no charge sharing)

A	B	C	D	E	G	H	K
1	1	1	X	1	1	1	X

first evaluate (no charge sharing)

A	B	C	D	E	G	H	K
X	X	X	0	X	X	X	0